

Karen Mahrous

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EDUCATION

University of Toronto

Bachelor of Applied Science, Computer Engineering + PEY Co-op

Toronto, ON

Sep. 2024 – Apr. 2029

École Secondaire Catholique Sainte-Famille

Bilingual Ontario Secondary School Diploma

Mississauga, ON

Sep. 2020 – May. 2024

EXPERIENCE

Engineering Intern

Vicwest

May. 2025 – Aug. 2025

Burlington, ON

- Designed and **deployed an Azure ML image detection** model for barcode and QR code recognition in warehouse shipments, achieving **80% detection accuracy**
- Created** detailed **CAD profiles** for roof claddings using AutoCAD, which were **published** on the company website and helped **speed up** the process of **releasing new products online**
- Built** an **automated reminder** and **license-tracking system** using **Excel** integrated with **Power Automate** that **eliminated manual follow-ups** and **ensured 100% on-time project updates**

Part-time Software Engineer Intern

Kingspan

Oct. 2025 – Present

Remote

- Continuing** from summer internship to **further develop** and **test** the **machine learning model using Azure ML**, **enhancing automation** and **classification accuracy**

Pharmacy Assistant, Coop student

Vibrin Drug Mart

Mar. 2024 – Jun. 2024

Mississauga, ON

- Assisted pharmacists in preparing and dispensing prescriptions, **improving efficiency** by streamlining workflows
- Conducted inventory checks and **ensured stock accuracy** during the co-op term
- Supported patient service by organizing prescriptions and ensuring timely medication pickup

PROJECTS

Interactive Map Application (ECE297) | University of Toronto

Jan. 2026 – Present

- Developed the **front-end and UI** of an interactive mapping application in **C++**, implementing responsive map rendering, street/intersection selection, search, and dynamic visual highlighting
- Improved graphics performance by building **feature, street segment, intersection, POI, and transit caches**, enabling faster rendering, visibility culling, and zoom-based level-of-detail display
- Implemented user-facing features including **day/night mode**, **category filters**, **street name rendering**, and a live **weather card** using **libcurl** and WeatherAPI

Game Explainer AI

Jan. 2026 – present

- Designing and developing a full-stack web application using **React** and **FastAPI** to deliver AI-assisted explanations for board game rules.
- Integrating **Azure OpenAI** for natural-language generation and a relational database to store and retrieve pre-generated explanations, enabling fast, consistent user responses.

City of Toronto Short-Term Rentals project | University of Toronto

Jan. 2025 – Apr. 2025

- Collaborated with a team to develop a mapping and route optimization system for STR inspections
- Designed an interactive GIS-based platform integrating automated logbooks, optimized routing (Dijkstra's/A*) to handle inspections efficiently

Leader, High School Earth day Initiative | École Secondaire Catholique Sainte-Famille

Sep. 2023 – Jun. 2024

- Spearheaded the school's Earth Day initiative in collaboration with the Best Buddy Club, expanding **reach and impact**
- Designed** event logos and **launched a website** featuring merchandise sales, **raising significant funds** for the World Wide Fund for Nature (WWF)

HONOURS AND AWARDS

Academic Excellence Award <i>École Secondaire Catholique Sainte-Famille</i>	2024
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SKILLS

Technical Skills & Tools: Azure, C, C++, Python, JavaScript, ExpressJS, Git, AutoCad, Excel, Power Automate Professional Skills: Leadership, Team Collaboration, Public Speaking, Problem Solving
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